Prayas JEE AIR (2025)

Advanced

WEEKLY TEST 25

DURATION : 180 Minutes | **DATE : 29/12/2024** | **M.MARKS : 180**

Topics covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 54 questions. The maximum marks are **180**.

SECTION-1 (Maximum Marks: 24)

This section contains **EIGHT (08)** Questions.

- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from **0 TO 9, BOTH INCLUSIVE**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +3 If **ONLY** the correct integer is entered;
  Zero Marks : 0 In all other cases.
  Negative Marks : –1 In all other cases.

SECTION-2 (Maximum marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +4 ONLY if (all) the correct option(s) is(are) chosen;
  Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;
  Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
  Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : –2 In all other cases.

SECTION-3 (Maximum marks: 12)

- This section contains **FOUR (04)** Questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : –1 In all other cases.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. Do not fold or make any stray mark on the Answer Sheet (OMR).

Name of the Students (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ Invigilator's Signature _____

IMPORTANT CONSTANTS

Speed of light in free space,	:	$3.00 \times 10^{8}\ ms^{-1}$
Permeability of free space,	:	$4\pi \times 10^{-7}\ Hm^{-1}$
Permittivity of free space,	:	$8.85 \times 10^{-12}\ Fm^{-1}$
The Planck constant,	:	$6.63 \times 10^{-34}\ Js$
Rest mass of electron,	:	$9.1 \times 10^{-31}\ kg$
Rest mass of proton,	:	$1.67 \times 10^{-27}\ kg$
Molar gas constant,	:	$8.31\ JK^{-1}\ mol^{-1}$
The Avogadro constant,	:	$6.02 \times 10^{23}\ mol^{-1}$
The Boltzmann constant,	:	$1.38 \times 10^{-23}\ JK^{-1}$
Gravitational constant,	:	$6.67 \times 10^{-11}\ N\ m^2kg^{-2}$
Acceleration of free fall	:	$9.8\ ms^{-2}$
Rydberg Constant	:	$1.097 \times 10^{7}\ m^{-1}$
Atomic mass unit	:	$1.67 \times 10^{-27}\ kg$
Charge on proton	:	$1.6 \times 10^{-19}\ C$

IMPORTANT VALUES

$\sqrt{2}$ = 1.414

$\sqrt{3}$ = 1.732

$\sqrt{5}$ = 2.236

π = 3.142

e (Euler's constant) = 2.718

ln 10 = 2.303

$\log_{10}2$ = 0.3010

$\log_{10}3$= 0.4770

$\log_{10}7$ = 0.845

**** Use above values unless otherwise specified in a question.***

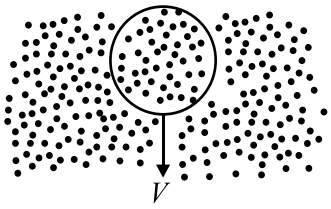
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PART-I (PHYSICS)

SECTION-1

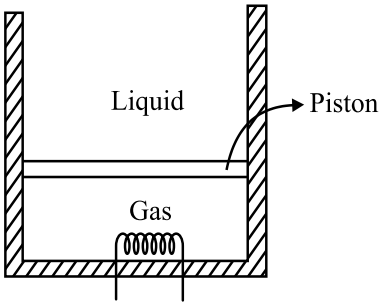
Single Digit Integer Type Questions:

1. A raindrop, initially of negligible mass, starts to descend from a large height in the absence of air. As it descends, it accumulates moisture (assumed at rest) surrounding it. This accumulation makes the raindrop grow in size and mass. Due to this accumulation, rate of change of instantaneous mass m (t) is proportional to the projected area of drop perpendicular to the velocity achieved and also proportional to the speed achieved by that moment. The proportionality constant, in *SI* units, is k. (Neglect the variation in earth's gravity)



When the drop attains terminal speed of 10 m/s, its mass is $\frac{4\pi}{3}$ grams and assume density of water to be 10^3 kg/m^3. Find the value of $3k$.

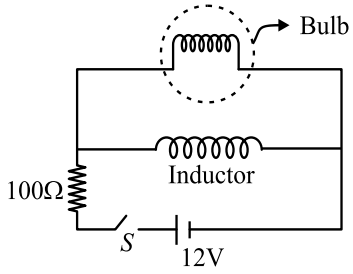
2. A cylinder has a height H and contains a liquid of density ρ (filled upto a depth of $2H/3$) and a monoatomic gas (at initial temperature T_0) separated by a thin massless piston of cross-sectional area A:



Now the gas is slowly heated using a coil and the expanding gas starts pushing the liquid out of the container. Total heat transferred to the gas till all the liquid is pushed out of the cylinder is $\frac{x}{9}\rho g H^2 A$. Find $\frac{x}{2}$. (Take atmospheric pressure $P_0 = \rho g H$) (Assume all surface to be insulating) [All surface to be frictionless]

3. Rate of change of acceleration with respect to time is sometimes given the name 'jerk'. The magnitude of jerk needed for a car to keep moving uniformly with a speed of 36 km/h in a circle of radius 10 m is x m/s^3. Find $\frac{x}{2}$.

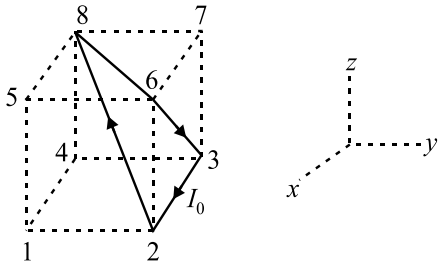
4. A bulb (purely resistive of resistance 100 Ω) is connected in a circuit using a *DC* battery of 12V as shown:



The inductor has a self inductance of 0.5 mH and self resistance of 100 Ω. The bulb stops glowing if the current through it is any less than 50 mA. Now, the switch is closed at $t = 0$. If the time interval (in μs), after closing the switch, for which the bulb will glow is x. Find $3x$. (Take $ln\ 2 = 0.69$).
(Answer should be Nearest Integer)

5. Consider a gaseous sample of Li^{2+} ions. All the ions (a total of 10^8) are in 3rd excited state. The radiations from this sample, when ions are transitioning to the ground state, are made to fall on a target sample of work function 95 eV. Assuming that the different possible journeys (of reaching the ground state from the initial state) are equilikely, the best estimate of the number of photoelectrons ejected from the target is $x \times 10^8$. Find $2x$. Take ionisation energy of an H atom to be 13.6 eV.

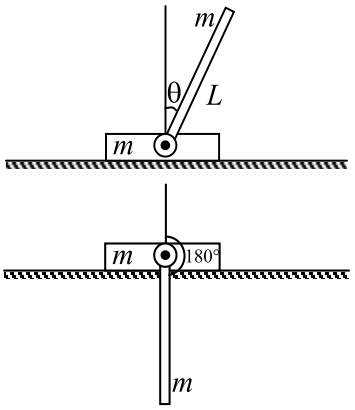
6. Imagine a cube of side 'a' whose corners have been labelled as 1, 2, 3, 4, 5, 6, 7, 8. Now, a conducting loop, carrying current I_0, is in the shape of a triangle whose vertices lie at points 2, 3 and 8.



The magnitude of magnetic field created by this loop at the centre of the imaginary cube is $n\sqrt{\frac{2}{3}}\frac{\mu_0 i}{\pi a}$. Find the value of $2n$.

7. If the velocity of light (c), gravitational constant (G) and Planck's constant (h) are chosen as fundamental units, then if the dimensional formula of mass in this new system is $\left[c^x G^y h^z\right]$. Find the value of $4(x + y + z)$

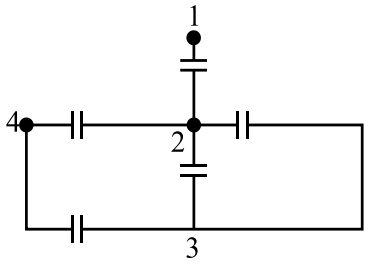
8. A rod of mass m and length $L = 50/3$ m is hinged in plank of same mass m. The plank is kept on a smooth horizontal surface and rod makes θ with the vertical. The system is released from $\theta = 0°$. If the velocity (in m/s) of plank when the rod makes $\theta = 180°$. ($g = 10$ m/s^2) is x. Find $x/2$



SECTION-2

One or More Than One Correct Type Questions:

9. Five identical capacitors (capacity C each) are connected to form a network as shown:



Some points have been marked on the network as 1, 2, 3 & 4. Across any pair out of these four points, we can connect an ideal battery and measure some parameters. Then, consider the following statements and choose the correct option(s).

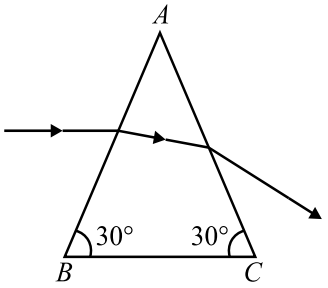
(A) If a DC battery is connected across points 1 and 2, then minimum charge on a capacitor is zero

(B) If a DC battery (voltage V) is connected across points 1 and 4, then minimum charge on a capacitor is $\frac{CV}{8}$

(C) Equivalent capacitance across points 2 and 3 is $\frac{3C}{2}$

(D) Equivalent capacitance across points 3 and 4 is $\frac{5C}{3}$

10. An isosceles prism ABC is made of a material of refractive index n. A ray of light, parallel to base BC, is incident on the face AB. A possible ray diagram, for some value of n, is shown:



Assume that the prism is surrounded by vacuum. Now changing the material of the prism, thereby changing the refractive index (without changing the geometry of the prism), would result in different paths being taken by the incident ray. Appropriate value of n might result in ray emerging from face AC or undergoing total internal reflection at that face.

Then, consider the following statements and choose the correct option(s).

(A) For rays emerging out from AC, angle of deviation first increases, on increasing value of n, and then decreases

(B) For rays emerging out from face AC, maximum allowed value of n is $\sqrt{\frac{7-2\sqrt{3}}{3}}$.

(C) For rays emerging out from AC, maximum possible angle of deviation is 30°.

(D) For ray emerging out from face AC, maximum allowed value of n is $\frac{\sqrt{7}}{2}$.

11. A particle is acted upon by a force of constant magnitude which is always directed perpendicular to the velocity of the particle. The motion of the particle takes place in a plane. From the information given, we conclude that the particle

(A) has constant velocity.

(B) has constant acceleration.

(C) has constant kinetic energy.

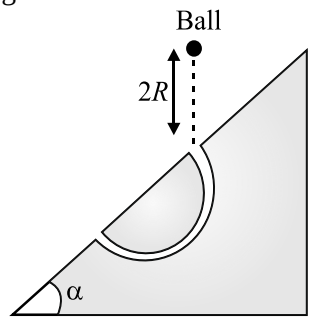
(D) moves in a circular path.

12. A cylindrical log of wood is floating in a large pool of water with its length normal to water's surface. The log has a radius r, mass m, length l and has density σ. If the log is depressed below its equilibrium depth d (but not beneath the surface of water) and then released, it executes harmonic oscillations with time period T. (Assume density of water to be ρ)

(A) $T = 2\pi\sqrt{\frac{m}{2\pi\rho rlg}}$ (B) $T = \frac{2\pi}{r}\sqrt{\frac{m}{\pi\rho g}}$

(C) $d = \frac{m}{2\pi\rho\, gl}$ (D) $d = \frac{m}{\pi r^2\rho}$

13. There exists a smooth semicircular groove in a rigid, light inclined wedge of inclination $\alpha = 60°$. The groove has a radius R and the wedge is attached fixed to a heavy scale at bottom which can measure the contact force between itself and the wedge. Now, a small smooth ball is dropped into the groove as shown:



Consider the collision of the ball (having mass m and released from rest) just as it enters in the groove to be completely inelastic. Then, choose the correct option(s) regarding the above situation.

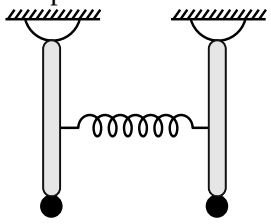
(A) Speed of ball just after collision is $\sqrt{gR}$

(B) The ball exits the groove with a speed $\sqrt{\left(2\sqrt{3}-1\right)gR}$

(C) Maximum reading of the scale (excluding the collision) is $\left(4+\sqrt{3}\right)mg$

(D) Minimum reading of the scale (as long as the ball is in the groove) happens just after the collision.

14. Some oscillating systems, instead of having one natural frequency, have two natural frequencies. Let the lower angular frequency be denoted by ω_L and the other one by ω_H. Consider one such system: Two physical pendulums, each comprising of a uniform rod of mass M and length L with small ball of mass M attached at one end, connected by a spring of spring constant k. The spring connects the mid-points of the rods and both the rods are hinged at their top points such that the whole arrangement is free to rotate in the vertical plane:



One of the natural frequencies results when both the rods are disturbed (together) by equal amounts in the same direction while the other natural frequency results when both the rods are disturbed (together) by equal amounts in opposite directions.

Now, consider the following statements and choose the correct option(s).

(A) Changing the stiffness of spring changes ω_H

(B) The ratio $\frac{\omega_H}{\omega_L}$ increases on increasing the dimension (L) of the system

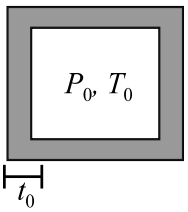
(C) $\omega_L = \sqrt{\frac{9}{8}.\frac{g}{L}}$

(D) $\omega_H = \sqrt{\frac{3}{8}\left[\frac{2k}{M}+\frac{3g}{L}\right]}$

SECTION-3

Single Correct Type Questions:

15. A rigid box contains one mole of an ideal gas of molar heat capacity at constant volume C. The walls of the box have total surface area A, thickness t_0 and thermal conductivity k. Initially the gas is at a temperature T_0 and pressure P_0. The temperature of the surrounding is $\frac{T_0}{2}$. The temperature of gas as a function of time, is



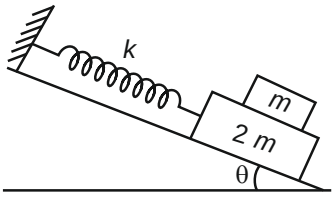
(A) $\frac{T_0}{2}\left[1-e^{\frac{-kAt}{Ct_0}}\right]$

(B) $\frac{T_0}{2}\left[1+e^{\frac{-kAt}{Ct_0}}\right]$

(C) $T_0\left[1-e^{\frac{-kAt}{Ct_0}}\right]$

(D) $T_0\left[1+e^{\frac{-kAt}{Ct_0}}\right]$

16. The coefficient of friction between block of mass m and $2m$ is $\mu = 2\tan\theta$. There is no friction between block of mass $2m$ and inclined plane. The elongation in spring at mean position is



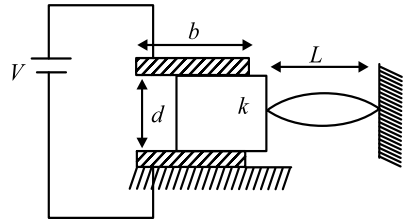
(A) $mg\sin\theta\frac{2}{k}$

(B) $\frac{mg\sin\theta}{k}$

(C) $\frac{3mg\sin\theta}{k}$

(D) $\frac{2mg\sin\theta}{k}$

17. A dielectric slab is attached to a string of mass per unit length μ, whose other end is fixed to a wall. Capacitor has square plates of side b and separation between the plates is d. Find the fundamental frequency of vibration of the string. (Dielectric slab remains in equilibrium and battery voltage is V)



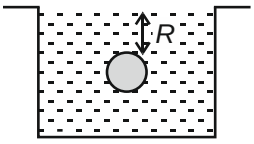
(A) $\frac{1}{2L}\sqrt{\frac{\epsilon_0 bV^2(2k-1)}{\mu d}}$

(B) $\frac{1}{2L}\sqrt{\frac{\epsilon_0 bV^2(2k+1)}{\mu d}}$

(C) $\frac{1}{2L}\sqrt{\frac{\epsilon_0 bV^2(k+1)}{2d\mu}}$

(D) $\frac{1}{2L}\sqrt{\frac{\epsilon_0 bV^2(k-1)}{2d\mu}}$

18. A sphere of radius R is submerged in a liquid of density ρ. The force exerted by liquid on upper half of sphere at the position shown in figure, is (assume atmospheric pressure to be zero)



(A) $\frac{3}{2}\pi R^3\rho g$ (B) $2\pi R^3\rho g$

(C) $\frac{1}{3}\pi R^3\rho g$ (D) $\frac{4}{3}\pi R^3\rho g$

PART-II (CHEMISTRY)

SECTION-1

Single Digit Integer Type Questions:

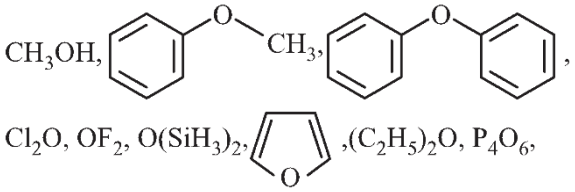
19. If the value of m can take values from $-(\ell+1)$ to $+(\ell+1)$ including zero then what will be the number of electrons in configuration of Cr having principal quantum number 3, keeping everything else intact?
[Atomic Number of Cr = 24]

20. Consider the following complexes:

Compound	No. of stereoisomers
$[Co(NH_3)_3Cl_3]$	x
$[Pt(Cl)(Br)(H_2O)_2]$	y
$[Co(C_2O_4)_3]^{3-}$	z

The value of (x + y + z) is _____.

21. How many of the following compounds have X — O — X (or Y) bond angle more than the H—O—H bond angle in water?



22. One mole of 1,2-Dibromopropane on treatment with x moles of $NaNH_2$ followed by treatment with ethyl bromide gave 2-pentyne. The value of x is _____.

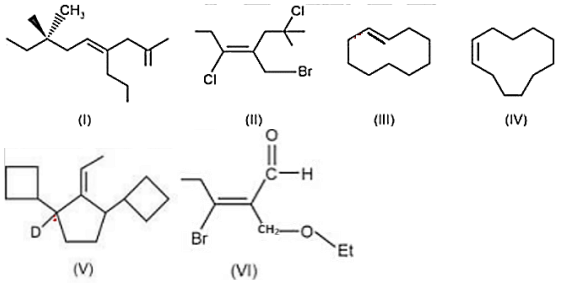
23. Two alkali metals A and B form bicarbonates. In the crystal structure of $AHCO_3$, HCO_3^- ions are linked via hydrogen bonding into an infinite chain while $BHCO_3$ forms a dimeric anion. If x and y are the atomic number of A and B respectively. Calculate the value of |x – y|.

24. Consider the following reaction sequence along with % yield.

$$Ph-CHO \xrightarrow[(ii)H_3O^+]{(i)MeMgBr/Et_2O} X \xrightarrow[50\%]{\text{Jones reagent}} Y \xrightarrow[80\%]{PhCHO/^-OH,\Delta} Z$$
80%

If the reaction is started with 1 mol of PhCHO and other reagents are sufficiently present. How much Z will be formed in decagrams? (Nearest integer)

25. How many of the following molecules are E-isomer?



26. Following radicals
Hg^{2+}, Pb^{2+}, As^{3+}, Bi^{3+}, Sb^{3+}, Sn^{2+}, Cu^{2+}, Cd^{2+}, Sn^{+4} give precipitate on treatment with H_2S in presence of dil. HCl. To these precipitates yellow ammonium sulphide solution is added and warmed up to 60° C with shaking. Precipitates of some of these radicals do not dissolve. How many of such radicals are there?

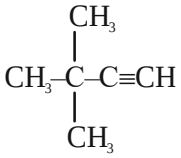
SECTION-2

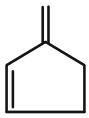
One or More Than One Correct Type Questions:

27. Which of the following is/are correct option for (P) in the given reaction sequence

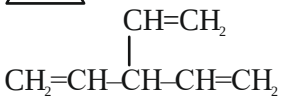
(P) Unsaturated Hydrocarbon $\xrightarrow[\text{(excess)}]{H_2/Hi}$ (Q)

Saturated Hydrocarbon $\xrightarrow{Cl_2/hv}$ 3-monochloro products.

(A) 

(B) 

(C) 

(D) 

28. At 25°C and 1 atm pressure $PCl_5(g)$ dissociates into $PCl_3(g)$ and $Cl_2(g)$. If it is 30% dissociated at given conditions. Mark the correct options.

(A) Value of $K_p = \frac{9}{91}$

(B) 8.95 ml of mixture will be diffused in the same time 20 ml of O_2 is diffused (Under similar conditions)

(C) $(\text{Molecular mass})_{mix} \simeq 160$

(D) Value of $K_p = \frac{27}{81}$

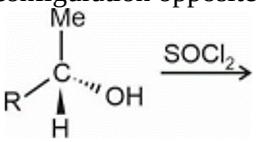
29. Consider the complex

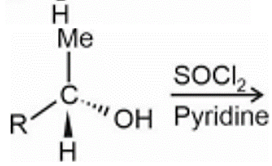
P: $[ML_6]^{x-}$ has crystal field stabilising energy $= -1.2\Delta_0$

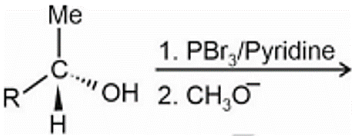
P is paramagnetic in nature on the basis of CFT. Select the correct combination(s) of metal (M) and (L) [L is monodentate ligand, x may be +ve or –ve integer]

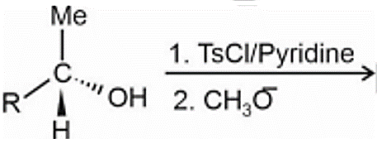
	M	**L**
(A)	Cr^{3+}	H_2O
(B)	Fe^{3+}	CN^-
(C)	Ni^{2+}	NH_3
(D)	Ni^{2+}	H_2O

30. In which of the following reactions, the product has configuration opposite to that of the reactant?

(A) 

(B) 

(C) 

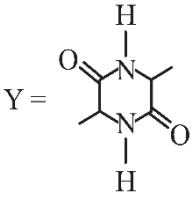
(D) 

31. $CH_3-CH_2-COOH \xrightarrow[\text{Red P}]{Br_2} \xrightarrow[\text{Alc.}]{NH_3}$

$X \xrightarrow{H^+/\Delta} Y$

Choose the correct option(s)

(A) Y gives three isomers

(B) Y = 

(C) $X = CH_3-\underset{\underset{O}{\|}}{C}-CH_2-NH_2$

(D) $CH_3-\underset{\underset{COOH}{|}}{CH}-NH_2 = X$

32. When an equimolar mixture of Cu_2S and CuS is titrated with $Ba(MnO_4)_2$ in acidic medium the final products contain Cu^{2+}, SO_2 and Mn^{2+}. If the molecular weight of Cu_2S, CuS and $Ba(MnO_4)_2$ are M_1, M_2 and M_3 respectively then:

(A) Equivalent weight of Cu_2S is $\frac{M_1}{8}$

(B) Equivalent weight of CuS is $\frac{M_2}{6}$

(C) Equivalent weight of $Ba(MnO_4)_2$ is $\frac{M_3}{5}$

(D) Cu_2S and CuS both have same equivalent in mixture

SECTION-3

Single Correct Type Questions:

33. How could one prepare 3,5-dibromophenol from para-nitroaniline ?

Note : $HNO_2 = NaNO_2 + 10\%H_2SO_4$

(A) (i) $2Br_2$ in ether (ii) HNO_2 0ºC (iii) H_3PO_2 (iv) $3H_2$ & Pt or Ni catalyst (v) HNO_2 0ºC, then heat

(B) (i) $3H_2$ & Pt or Ni catalyst (ii) HNO_2 0ºC (iii) excess Cu_2Br_2 + HBr (iv) KOH & heat

(C) (i) HNO_2 0 ºC, then heat (ii) $2Br_2$ in ether (iii) $3H_2$ & Pt or Ni catalyst (iv) HNO_2 0ºC (v) Cu_2Br_2 + HBr

(D) (i) $2Br_2$ in ether (ii) HNO_2 0ºC (iii) Cu_2Br_2 (iv) $3H_2$ & Pt or Ni catalyst (v) HNO_2 0ºC, then heat

34. The Ksp of $Pb_3(AsO_4)_2$ is 1.08×10^{-28} at 27°C. Find $E°_{cell}$ for the reaction:

$Pb_3(AsO_4)_{2(s)} + 6e^- \rightleftharpoons 3Pb_{(s)} + 2AsO_4^{-3}$ if

$E^{\circ}_{Pb^{2+}/Pb} = -0.13\text{Volt}$

$\left\{\text{Given}: \frac{2.303RT}{F} = 0.06, \quad \log 2 = 0.30\right\}$

(A) + 0.18 V (B) – 0.19 V

(C) – 0.016 V (D) – 0.04 V

35. A monoatomic ideal gas undergoes a process in which the ratio of P to V at any instant is constant and equals to 1. The molar heat capacity of the gas:

(A) 0 (B) 2R

(C) $\frac{3R}{2}$ (D) $\frac{5R}{2}$

36. In winter, the normal temperature in Kullu valley was found to be −11°C. Is a 28% (by mass) aqueous solution of ethylene glycol suitable for a car radiator ? K_f for water = 1.86K kg mol^{-1}.

(A) Yes

(B) No

(C) can't predict

(D) 18% (by mass) solution is also suitable

PART-III (MATHEMATICS)

SECTION-1

Single Digit Integer Type Questions:

37. Let $H_n : \frac{x^2}{1+n} - \frac{y^2}{3+n} = 1,\ n \in \mathbb{N}$. Let k be the smallest even value of n such that the eccentricity of H_k is a rational number. If l is the length of the latus rectum of H_k, then sum of digits of $21l$ is equal to

38. Let line L passes through the origin and be perpendicular to the lines

$L_1 : \vec{r} = (\hat{i} - 11\hat{j} - 7\hat{k}) + \lambda(\hat{i} + 2\hat{j} + 3\hat{k}), \lambda \in \mathbb{R}$ and

$L_2 : \vec{r} = (-\hat{i} + \hat{k}) + \mu(2\hat{i} + 2\hat{j} + \hat{k}), \mu \in \mathbb{R}$. If P is the point of intersection of L and L_1 and, $Q(\alpha,\beta,\gamma)$ is the foot of perpendicular from P on L_2, then $9(\alpha + \beta + \gamma)$ is equal to ____.

39. If $I_1 = \int_0^1 \frac{x^{7/2}(1-x)^{9/2}}{30}dx$ and $I_2 = \int_0^1 \frac{x^{7/2}(1-x)^{9/2}}{(x+5)^{10}}dx$ and $\frac{I_1}{I_2} = 5a^3\sqrt{a}$ where $a \in N$, Then the value of $\frac{a}{10}$ is

40. If the area of region $\left\{(x,y) \in R^2 : y^2 \geq 4x, |y| \leq \frac{x}{2} + 2\right\}$ is Δ, then the value of $\frac{[\Delta]}{2}$ is (where $[\cdot]$ denotes greatest integer function)

41. $4\lim_{n\to\infty}\left[\frac{1}{n} + \frac{n}{(n+1)^2} + \frac{n}{(n+2)^2} + \ldots.. + \frac{n}{(2n-1)^2}\right]$ to is equal to:

42. If $f(x) + f(y) + f(xy) = 2 + f(x) \cdot f(y)$, for all real values of x and y and $f(x)$ is a polynomial function with $f(4) = 17$ and $f(1) \neq 1$, then the value of $\frac{f(7)}{10}$ is

43. If the sum of $\sqrt{1+\frac{1}{1^2}+\frac{1}{2^2}} + \sqrt{1+\frac{1}{2^2}+\frac{1}{3^2}} + \sqrt{1+\frac{1}{3^2}+\frac{1}{4^2}} + \ldots\ldots + \sqrt{1+\frac{1}{1999^2}+\frac{1}{2000^2}}$ is equal to $\left(n - \frac{1}{n}\right), n \in N$, then the sum of the digits the of n is

44. Let z be a complex number satisfying $|z - 3| \leq |z - 1|$, $|z - 3| \leq |z - 5|$, $|z - i| < |z + i|$ and $|z - i| \leq |z - 5i|$. Then the area of region in which z lies is A square units, where A is

SECTION-2

One or More Than One Correct Type Questions:

45. Given that a,b,c are positive distinct real numbers such that quadratic expressions $ax^2 + bx + c, bx^2 + cx + a$ and $cx^2 + ax + b$ are always non-negative. Then the expression $\frac{a^2+b^2+c^2}{ab+bc+ca}$ can never lie in :

(A) $(-\infty, 2]$ (B) $(-\infty, 1]$

(C) (2, 4) (D) $[4, \infty)$

46. Identify the correct statement(s).

(A) Number of naughts standing at the end of $\lfloor 125$ is 30

(B) A telegraph has 10arms an each arm is capable of 9 distinct positions excluding the position of rest. The number of signals that can be transmitted is $10^{10}-1$

(C) Number greater than 4 lacs which can be formed by using only the digits 0, 2, 2, 4, 4 and 5 is 90

(D) In a table tennis tournament, each player plays with every other player. If the number of games played is 5050 then the number of players in the tournament is 100

47. The set of values of x for which $f(x) = \ln(x^2 - 7x + 10)$ is always increasing is

(A) $\left(2, \frac{7}{2}\right) \cup (5, \infty)$ (B) $(5, \infty)$

(C) ϕ (D) R

48. The solution of $\left(\frac{xdx+ydy}{xdy-ydx}\right) = \sqrt{\frac{a^2-x^2-y^2}{x^2+y^2}}$ is

(A) $\sqrt{\left(x^2+y^2\right)} = a\sin\left\{\left(\tan^{-1} y/x\right) + \text{constant}\right\}$

(B) $\sqrt{\left(x^2+y^2\right)} = a\cos\left\{\left(\tan^{-1} y/x\right) + \text{constant}\right\}$

(C) $\sqrt{\left(x^2+y^2\right)} = a\tan\left\{\left(\sin^{-1} y/x\right) + \text{constant}\right\}$

(D) $y = x\tan\left\{\text{constant} + \sin^{-1}\frac{1}{a}\sqrt{\left(x^2+y^2\right)}\right\}$

49. Let X and Y be two events such that $P\left(\frac{X}{Y}\right) = \frac{1}{2}, P\left(\frac{Y}{X}\right) = \frac{1}{3}$ and $P(X \cap Y) = \frac{1}{6}$. then

(A) $P(X \cup Y) = \frac{2}{3}$

(B) $P(X^C \cup Y^C) = \frac{1}{3}$

(C) X and Y are independent events

(D) $P(X^C \cap Y) = \frac{1}{3}$

50. If $M = \begin{bmatrix} 0 & 1 & a \\ 1 & 2 & 3 \\ 3 & b & 1 \end{bmatrix}$ and $\text{adj}(M) = \begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{bmatrix}$ where a and b are real numbers then

(A) $a = 2$

(B) $b = 1$

(C) $a = 1$

(D) $b = 2$

SECTION-3

Single Correct Type Questions:

51. The sum ${}^nC_0 + {}^nC_1 - {}^nC_2 - {}^nC_3 + {}^nC_4 + {}^nC_5 \ldots$ is equal to

(A) $2^{n/2}\left(\cos\frac{n\pi}{4} + \sin\frac{n\pi}{4}\right)$

(B) $2^{n/2}\left(\cos\frac{n\pi}{4} - \sin\frac{n\pi}{4}\right)$

(C) $2^{n/2}\cos\frac{n\pi}{4}$

(D) $2^{n/2}\sin\frac{n\pi}{4}$

52. Locus of image of (4, 6) about the family of lines $a(3x - 4y + 5) + b(x + y - 3) = 0$ is a circle of radius

(A) 2

(B) 3

(C) 5

(D) 7

53. The expression $\cos^2(\alpha + \beta) + \cos^2(\alpha - \beta) - \cos 2\alpha \cos\beta$ is independent of

(A) α only

(B) β only

(C) Both α and β

(D) None of these

54. The value of $\lim_{x\to\infty}\left(\sin\frac{1}{x} + \cos\frac{1}{x}\right)^x$ is

(A) e^2

(B) $e^{-1/2}$

(C) e

(D) e^{-1}



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